Harrison He

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EDUCATION

University of Illinois Urbana-Champaign Expected May 2030
Bachelor of Science in Mathematics (transferring into Mathematics & Computer Science)
Coursework: MATH 241 Calculus III · MATH 257 Linear Algebra · CS 124 Intro to Computer Science I · MATH 231 by proficiency exam
Activities: GTO Illini (poker strategy) · Financial Engineering Club · Illinois Data Science Club

St John's College, Johannesburg Graduated 2025
Cambridge A-Levels: A Mathematics, A* Further Mathematics, A Physics, A Chinese, B Chemistry*

PROJECTS & EXPERIENCE

Founder & Engineer — 店小力 Dianxiaoli, production AI sales agent for small merchants Apr 2026 – Present · Remote
*Incubated by **Du Xiaoman Financial** (Baidu's fintech arm), 小满摘星计划, since Sep 2026 · write-up: harrisonhe.com/dianxiaoli.html*

- Conceived, built and operate, alone, an autonomous sales agent inside WeChat Customer Service and WhatsApp: quotes from the merchant's catalogue, negotiates, records orders, handles payment screenshots, escalates edge cases to the owner. **Live 24/7 for 3 paying merchants on two continents**; first deployment running since 2025.
- **Safety in code, not prompts:** no invented prices, no stock counts, no cross-customer prices, never confirm a payment — a post-model price guard voids any non-catalogue price, payments move only by owner action on an append-only ledger, and a nightly 9-check audit of production transcripts is itself verified by mutation tests.
- **Platform (Sep 2026):** multi-tenant core serving many stores per deployment; a staff role with per-field visibility, price-change approvals and enquiry tickets; four industry packs and 14 conversation skill packs; a consent-gated 17-field merchant-health export consumed by a fintech partner.
- **Engineering:** deterministic-first pipeline (rule engine first, LLM only on the unmatched path); one model call per turn emitting tagged actions validated by a deterministic executor; async owner notifications (P95 < 3 s); **18 suites / 1,055 checks** gate every release, including word-for-word golden tests against the previous release; ≈¥100 per store per month; deployment 3 weeks → 2–3 days.
- Refactored the original production codebase into 9 reusable modules; wrote the roadmap, partner-facing product and API docs and an 18-page onboarding handbook; run a two-agent spec → implement → review workflow for my own engineering.
- **Du Xiaoman incubation (Sep 2026 – present):** selected into the programme to build the standardised, multi-tenant edition for Du Xiaoman's small-business merchants, with a joint company planned once it reaches scale; the programme contributes two assigned contributors and pilot access to its merchant base. Own product and engineering; set the roadmap with their team; designed the merchant-data interface for their risk models; delivered the multi-tenant core and industry packs three weeks ahead of the agreed schedule.

Incoming Intern — Allan Gray, Johannesburg Dec 2026

- Confirmed internship at one of South Africa's largest independent asset managers (role and dates to be confirmed).

AI Study Tutor — handwritten past-paper grading 2025 – Present

- Built an LLM pipeline (vision OCR → grading model → step-by-step explanation and new-problem generation) that marks a handwritten answer by logical validity, accepting any correct method; FastAPI on Railway, Next.js on Vercel.
- **Hand-audited the model against a full paper: 9 of 15 AI marks were false negatives, all in one direction** — now a 13-case regression suite gating every change.

Cambridge Research Project — regression analysis of cryptocurrency price series St John's College

Actuarial Job Shadows — Discovery; Argen Actuarial Solutions Apr – May 2024 · Johannesburg

- Observed pricing models, risk assessment and actuarial valuations.

ACHIEVEMENTS

- **South African Mathematics Olympiad:** 4th (2021), 7th (2022), Top 50 (2023), Top 20 (2024, 2025) nationally; South Africa IMO selection training camp, Stellenbosch University, 2021–2024.
- **University of the Witwatersrand Mathematics Competition:** Top 10, 2023–2025. South Africa Team Maths Competition: 2nd place 2021–2024, 4th 2025. Estonian Maths Open: Top 20–50 within South Africa.
- South African Informatics Olympiad: first round passed 2023, 2024. Qualified Professional Level 1 Go player (China). 100+ hours volunteer tutoring in high-school mathematics and olympiad preparation, 2022–2025.

SKILLS

Programming: Python (FastAPI, pandas), JavaScript / HTML / CSS (Next.js), Java, Git, Playwright; Railway and Vercel operations.
Languages: Mandarin and English, both fluent.

AI & data engineering: LLM agent design with DeepSeek, Anthropic and OpenAI APIs — tagged-action output, deterministic executors, rule-engine fallback, background workers, multi-tenant routing, role-based field visibility; evaluation gates, golden and mutation tests, transcript audits; consent-gated data export; WeChat and WhatsApp platform integration.